A.L.T.R.U.S – Adaptive Life-support & Therapeutic Robotic Unit System

Project ID: 25-26j-437

Project Proposal

Semasinghe S.N.S.B.K.K. – IT22084668

Supervised by: Mrs. Hansi De Silva

Co-supervised by Mr. Ashvinda Iddamalgoda

B.Sc. (Hons) Degree in Information Technology

Department of Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

July 2025

DECLARATION

We are confirming that the work given in this proposal is one of our own volition. The same work has not been given, entirely or partially, for academic credit at any other university or institution. We have also made sure that all sources of information and ideas from others have been clearly cited and appropriately acknowledged in the document.

Name	Student ID	Signature
Semasinghe S.N.S.B.K.K	IT22084668	

The above candidates are researching the undergraduate Dissertation under my supervision.

Name of supervisor: Ms. Hansi De Silva

Name of co-supervisor: Mr. Ashvinda Iddamalgoda

Signature of supervisor: Date: 2025/08/

Signature of co-supervisor: Date: 2025/08/

ABSTRACT

This project presents an intelligent and adaptive navigation and interaction system for home assistive robots, designed to operate reliably in dynamic domestic environments. It combines static Simultaneous Localization and Mapping (SLAM) with real-time obstacle detection via LIDAR, enabling dynamic path replanning in response to environmental changes such as furniture rearrangement or the appearance of new obstacles. This hybrid mapping approach allows for safe, autonomous navigation without the need for full remapping. To support natural human-robot interaction, the system incorporates simple voice command recognition and gesture-based control. These intuitive interfaces are especially beneficial for elderly or physically challenged users. The solution is built on the Robot Operating System 2 (ROS2) framework, leveraging its modular and scalable architecture to ensure real-time performance. A central decision engine processes input from interaction modules and coordinates corresponding navigation behaviors. Additional capabilities include autonomous docking, environmental change detection, and context-aware behavior. By integrating adaptive navigation with user-friendly interaction methods, this system enhances the autonomy, safety, and practicality of assistive robots in real-world home settings. Its design is well-suited for applications in smart homes, elderly care centers, and independent living environments.

Keywords:

Adaptive Navigation, Dynamic SLAM, Human-Robot Interaction, Voice Command, Gesture Recognition, Assistive Robotics, ROS2

TABLE OF CONTENTS

DECLARATION 2
ABSTRACT 3
TABLE OF CONTENTS 4
LIST OF FIGURES 6
LIST OF ABBREVIATIONS 7
1. INTRODUCTION 8
1.1 BACKGROUND 8
1.2 LITERATURE SURVEY 10
1.2.1 Personalized Anomaly Detection in wearable Devices 10
1.2.2 Zero-Shot and Domain Adapted ECG monitoring 10
1.2.3 Fall Detection in Wearable Health Monitoring 11
1.3 RESEARCH GAP 11
1.3.1 LSTM-Based ECG Classification for Continuous Monitoring on Personal Wearable Devices 11
1.3.2 ECG-TCN: Wearable Cardiac Arrhythmia Detection using Temporal Convolutional Network-based ECG-TCN 11
1.3.3 A Zero-Shot Personalized ECG Arrhythmia Monitoring System 12
1.3.4 A Zero-Shot Personalized ECG Arrhythmia Monitoring System 12
1.3.5 A Zero-Shot Personalized ECG Arrhythmia Monitoring System 12
1.4 RESEARCH PROBLEM 13
2. RESEARCH OBJECTIVES 14
2.1 MAIN OBJECTIVE 14
2.2 SPECIFIC OBJECTIVES 14
2.2.1 Multi-Sensor Hardware Integration 14
3. METHODOLOGY 16
3.1 Requirements Analysis 17
3.2 Hardware Development for Multi-Sensor 17
3.3 Data Acquisition and Preprocessing 18
3.4 Machine Learning Model Development 18
3.5 System Integration with A.L.T.R.U.S. 19
3.6 Analysis and Performance Metrics 19

4. PROJECT REQUIREMENTS 20
4.1 Functional requirements 20
4.2 Non-Functional requirements 21
5. BUDGET DETAILS 22
6. CONCLUSION 23
References 24
APPENDICES 25
Appendix 1. Work Breakdown Structure (WBS) 25

LIST OF FIGURES

Figure 1.1.1Survey on most useful features in health emergency detection wristband 9
Figure 2.1.2 Survey on how likely they act on medical emergencies 9
Figure 3.3.1 System Overview Diagram..... 16
Figure 4.3.2 My component system diagram..... 17

LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ECG	Electrocardiogram
PPG	Photoplethysmography
SpO_2	Peripheral Capillary Oxygen Saturation
TCN	Temporal Convolutional Network
LSTM	Long Short-Term Memory
BLE	Bluetooth Low Energy
ROS 2	Robot Operating System 2

1. INTRODUCTION

1.1 BACKGROUND

The integration of wearable health-monitoring systems into assistive robotics has gained significant attention in recent years with advanced miniaturized sensors, wireless communications and machine learning for real-time anomaly detection. Assistive robots are increasingly used to support elderly individuals, patients with chronic illnesses and people who require long term continuous health supervision.

While existing robots excel in tasks such as reminders, companionship and basic mobility assistance, their effectiveness is limited when it comes to continuous and autonomous health monitoring without manual intervention.

A fundamental requirement for ensuring safety in home-based healthcare is the ability to detect health emergencies as soon as they occur or detect emergencies before they get critical using symptoms. Conventional wearable devices such as fitness bands and smartwatches offer basic tracking features like heart rate and step count but are not specifically designed for clinical grade monitoring that involves doctor for critical situations or rapid integration with robotic emergency response systems. Most existing solutions either rely on manual data checks or require devices that lack user comfort and mobility..

The problem is especially critical in detecting conditions such as heart attacks, tachycardia, bradycardia, cardiac arrest, falls and fever. These all should have anomaly detection and immediate actions. Delayed detections, even by a few minutes can result in serve medical consequences or death. In many domestic situations elderly person may be alone and unable to call help, So making automatic detection and alert systems essential.

To address these challenges, the proposed wristband component of A.L.T.R.U.S. combines multiple sensors including heart rate, SpO_2, skin temperature and accelerometer & gyroscope to collect vital data continuously. The data is processed using onboard and ROS2 compatible algorithms capable of detecting abnormalities in real time and communicating directly with the assistive robot for autonomous response such as alerting caregivers, direct calling doctor, contact emergency services or instruct first aid steps.

User Feedback to Support the Research

To validates the need for such a robot , a user survey was conducted with healthcare professionals, caregivers and potential end users.

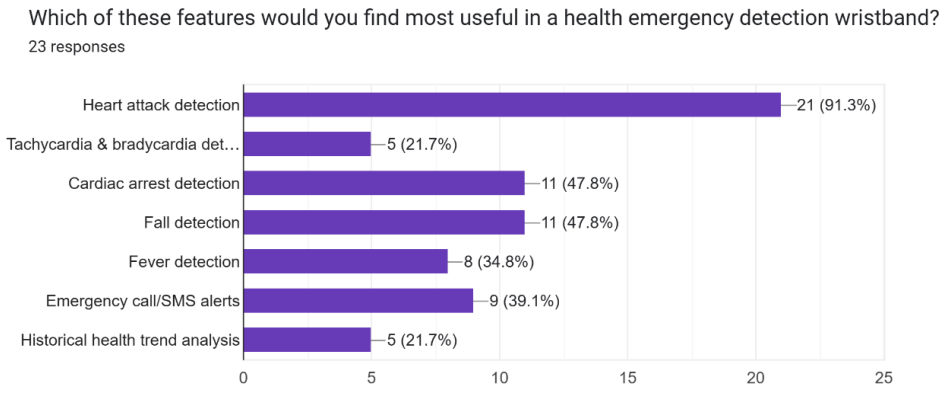


Figure 1.1.1Survey on most useful features in health emergency detection wristband

Figure 1.1.1. illustrate survey findings on the which features that would be include in emergency detection wristband. Most common(91.3%) feature that users expecting is heart attack detection. There are two similar percents of responses on two features and that is the next common (47.8%) request are cardiac arrest detection and fall detection. Survey has the lowest (21.7%) response on Tachycardia & bradycardia detection.

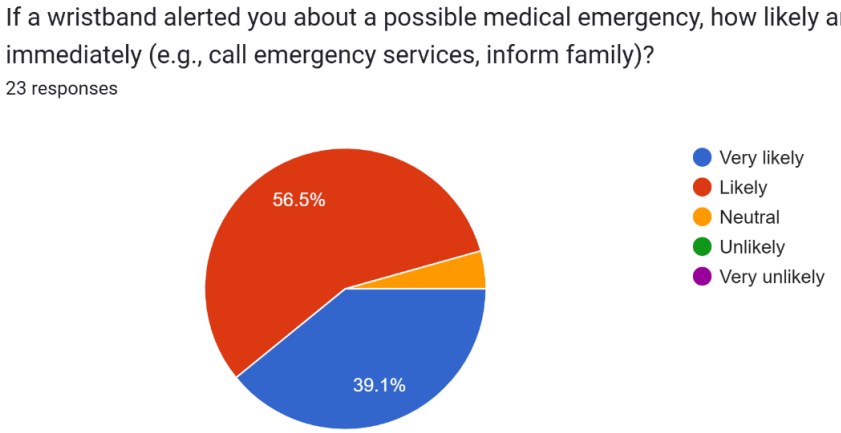


Figure 2.1.2 Survey on how likely they act on medical emergencies

Figure 1.1.2. indicates that people are generally willing to take immediate actions if a wristband alerts them about a possible medical emergency. About 39.1% are very likely to response right away, While 56.5% are likely to do so and only small portion remain neutral.

Conclusion

The survey findings confirm that there is a significant unmet need for an intelligent, comfortable and integrated health monitoring system in the domestic assistive robot domain. The proposed wristband for A.L.T.R.U.S. addresses this by combining multiparameter sensing, real-time anomaly detection, connect doctor directly in critical emergencies.

By bridging the gap between wearable health technology and autonomous robotics, this solution enables proactive healthcare reducing response times in emergencies and improving trust in robotic caregiving systems. Implemented within a ROS2 framework, the system ensures scalability, modularity and real time performance without reliance on cloud-based processing, making it robust for real-world deployments.

1.2 LITERATURE SURVEY

This section provides an overview of the latest research and technology in terms of sensor fusion real-time health anomaly detection wrist band and their strengths and limitations compared to the system discussed here.

1.2.1 Personalized Anomaly Detection in wearable Devices

[1] Studies such as AI on the pulse : Real-Time Health Anomaly Detection with Wearable and Ambient Intelligence(Gabrielli et al., 2025) have demonstrated the potential of integrating wearable sensors and ambient intelligence for continuous monitoring. It utilizes patient specific baseline modeling and lightweight devices such as smartwatches for real time detection with significant performance gains in real-world home deployments. Although helpful these systems rely heavily on ongoing environmental sensing, which may limit portability in situations where ambient data is unavailable.

1.2.2 Zero-Shot and Domain Adapted ECG monitoring

[2] *A Personalized Zero-Shot ECG Arrhythmia Monitoring System* (Yamaç et al., 2024) paper introduces the method that called sparse representation-based domain adaptation to identify arrhythmias that does not require patient-specific anomaly data during model training. That achieves up to 98.2% accuracy on benchmark datasets. This approach is suitable for wearables and it also energy efficient. However as the problem that common wrist worn ECG implementation has, models performance can be influenced by sensor quality and signal noise.

1.2.3 Fall Detection in Wearable Health Monitoring

[3] The *CareFall: Automatic Fall Detection through Wearable Devices and AI Methods* study presents real time fall recognition using integrated Gyroscope and accelerometer sensor integrated into wearable wrist band. The system processes sensor data through hybrid AI pipeline combining feature extraction with supervises classifiers such as support vector machines(SVM) and random forests. CareFall demonstrate accuracy rate above 95% for hard falls while maintaining a low false alarm rate for daily activities like bending, sitting in semi-natural environments. However, its performance deteriorates when the wearable device is not worn on the optimal location on the body, and the system primarily concentrates on fall detection and not other essential health abnormalities. This one-functionality limits its applicability in multi-condition emergency alarm systems such as the one proposed within the A.L.T.R.U.S. wristband, which attempts to integrate fall detection with cardiac, breathing, and thermal abnormality monitoring in a single low-power system.

1.3 RESEARCH GAP

1.3.1 LSTM-Based ECG Classification for Continuous Monitoring on Personal Wearable Devices

This paper demonstrates that LSTM networks can be employed for continuous ECG classification in wearable systems to identify arrhythmia with low latency and correct results. It does not account for non-cardiac irregularities and multi-sensor information such as SpO_2, body temperature, or accelerometer signals. Not accounting for sensor fusion in this paper limits its application to find other emergencies (e.g., falls, fever) that are crucial in an overall health monitoring system.

1.3.2 ECG-TCN: Wearable Cardiac Arrhythmia Detection using Temporal Convolutional Network-based ECG-TCN

ECG-TCN is a superior arrhythmia detection model using temporal convolutional networks that presents an effective alternative to recurrent models. Though it performs superiorly in sequential ECG data processing, it remains a single-condition detection model. It lacks real-time multi-condition adaptability and does not encompass the power optimization techniques that would be needed to integrate it for long-term use in wearable applications in multi-sensor systems.

1.3.3 A Zero-Shot Personalized ECG Arrhythmia Monitoring System

Domain Adaptation from Sparse Representation to Energy-Efficient Abnormal Beat Detection for Realistic ECG Surveillance

This approach is better in terms of personalization and energy-efficient ECG monitoring through domain adaptation. However, its use is only for arrhythmia detection without any use of motion or temperature sensing. Without including multi-modal integration of data, it cannot provide an integrated emergency detection solution viable for home-based health robotics.

1.3.4 A Zero-Shot Personalized ECG Arrhythmia Monitoring System

CareFall gives robust fall detection from IMU readings using AI classifiers with high sensitivity in diverse scenarios. But it is restricted to falls alone and does not identify anomalies in vital signs. The lack of other medical monitoring capability makes it impossible to be an all-in-one wearable device for more than one emergency condition.

1.3.5 A Zero-Shot Personalized ECG Arrhythmia Monitoring System

It is a combination of ambient and wearable sensing for real-time anomaly detection. While it adds sense beyond single metrics, it is dependent on ambient sensors outside of the body, hence less portable and infrastructure-dependent. It is also lacking direct ECG-based cardiac event detection, reducing its capability to detect specific cardiovascular emergencies in a wearable-only application.

1.4 RESEARCH PROBLEM

As companion robots are increasingly being utilized in healthcare facilities to monitor, aid, and provide emotional support, their utility is limited by the absence of ongoing physiological data acquisition. Crucial health parameters such as heart rate, cardiac rhythm disturbances, fall incidents, and core body temperature often must be closely and constantly monitored.A capability that cannot always be guaranteed by immobile or semi-mobile robots.

Contactless measurement modalities embedded in robots, such as infrared thermography or vision-based facial expression analysis, are inherently dependent on optimal environmental conditions. Line-of-sight occlusion, varying light, user motion, or partial occlusion significantly reduce measurement validity. Furthermore, such systems cannot ensure continuous health monitoring when the user steps out of the robot's sensor range, enters another room, or is occluded by furniture and walls.

In real home settings, this lack of ongoing monitoring is a life-critical safety flaw. If a medical crisis, such as a heart attack, cardiac arrest, tachycardia, bradycardia, sudden fever, or fall, occurs outside the robot's direct line of vision, the system may not be in a position to detect the event at all. This calls for an immediate need for an efficient, non-intrusive, and wearable health monitoring system that will operate effectively in conjunction with the robot.

Denied a unified wearable capable of continuously collecting, processing, and transmitting biometric data in real time, companion robots remain half "blind" to the user's health status, severely curtailing their ability to launch timely, context-aware, and potentially life-saving interventions.

2. RESEARCH OBJECTIVES

2.1 MAIN OBJECTIVE

The primary aim of this study is to design and integrate a multi-sensor wearable wristband with the A.L.T.R.U.S. health robot, enabling round-the-clock, real-time monitoring of vital physiological and motion parameters. The system will surpass the limitations of current wearable and robotic health monitoring systems by incorporating ECG, heart rate, body temperature, and motion sensing into one platform. Through state-of-the-art machine learning–based anomaly detection, personalized calibration, and sophisticated sensor fusion techniques, the wristband will facilitate accurate detection of a range of emergency events, such as cardiac arrest, tachycardia, bradycardia, fever, and falls. By creating reliable, low-latency communication with the companion robot, the proposed solution will allow effective emergency response even when the user is beyond the robot's immediate sensing range, in addition to maintaining excellent performance under diverse real-world conditions.

2.2 SPECIFIC OBJECTIVES

2.2.1 Multi-Sensor Hardware Integration

Develop a wearable wristband that combines ECG, heart rate, body temperature, and motion sensors to address the limitations of single-sensor designs in existing health monitoring systems.

2.2.2 Hybrid Anomaly Detection Models

Utilize a machine learning pipeline that combines temporal models (e.g., TCN, LSTM) with personalized threshold adaptation for better detection accuracy of arrhythmia and fall events.

2.2.3 Robust Signal Preprocessing

Develop signal filtering and sensor fusion algorithms to minimize false positives from motion artifacts, ambient noise, and variable sensor contact.

2.2.4 Real-Time Data Transmission

Develop transparent, low-latency communication between the wristband and the A.L.T.R.U.S. robot to maintain continuous monitoring even when the user moves outside of the robot's line of sight.

2.2.5 Comprehensive Multi-Event Detection

Create a detection system capable of identifying numerous life-threatening health emergencies in a single platform, going beyond the event-specific focus of the majority of current research.

2.2.6 Realistic Scenario Validation

Test the system performance in varying environmental conditions, lighting, occlusions, user movements, and prolonged daily wear to ensure real-world applicability.

3. METHODOLOGY

The intended methodology is organized into five broad stages: Requirements Analysis, Hardware Development, Data Acquisition, Machine Learning Model Development, and System Integration with A.L.T.R.U.S. Each stage is intended to sequentially fill the defined research gaps and achieve the stated objectives.

OVERALL SYSTEM DIAGRAM

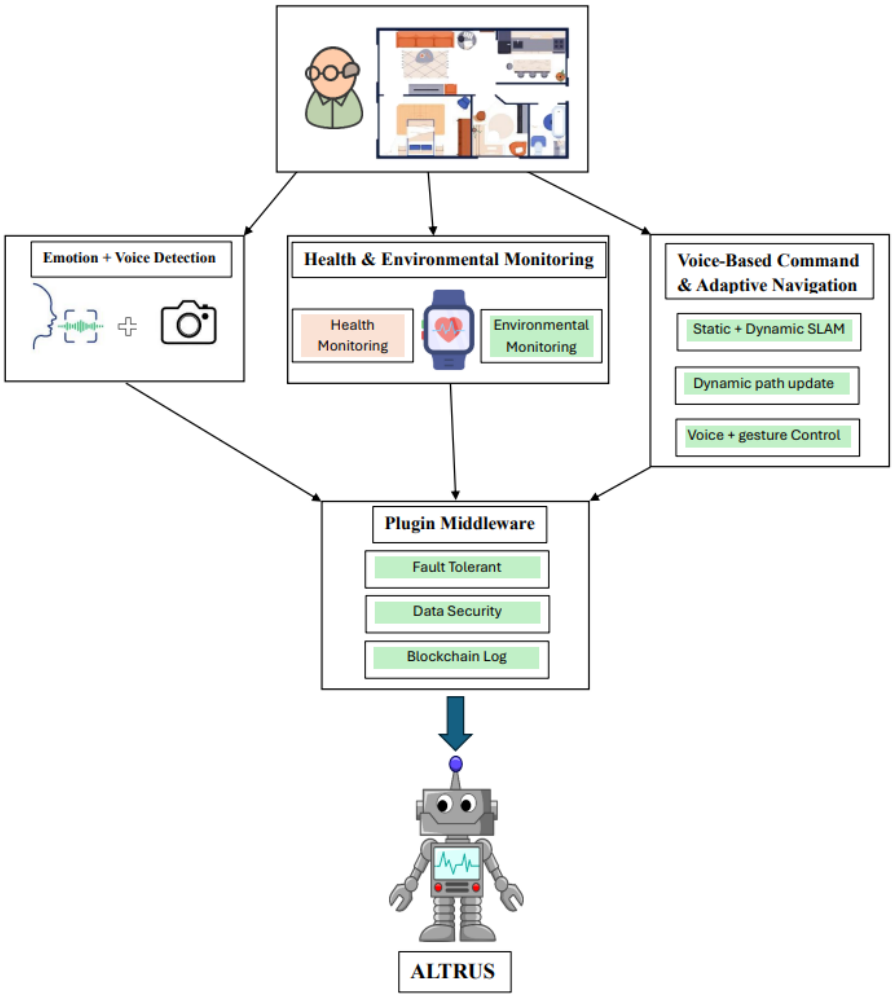


Figure 3.3.1 System Overview Diagram

MY COMPONENT SYSTEM DIAGRAM

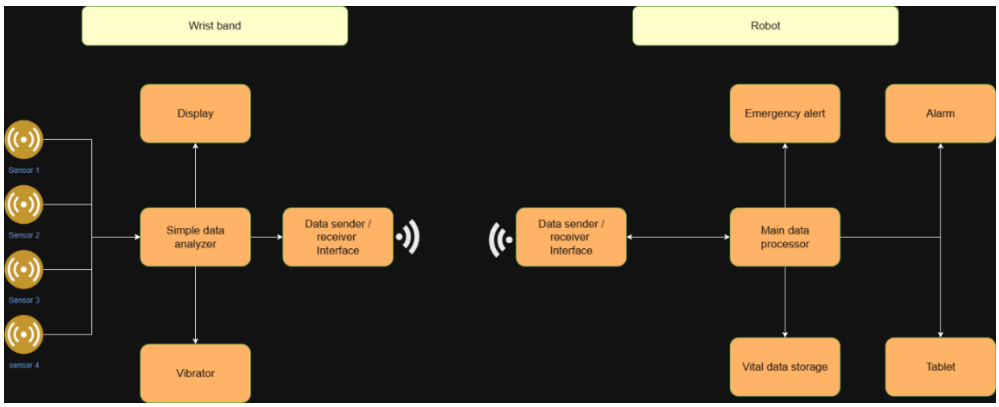


Figure 4.3.2 My component system diagram

3.1 Requirements Analysis

Conduct an extensive survey of contemporary wearable healthcare platforms and robotic surveillance methods to determine the final set of life signs and emergencies to be detected (cardiac arrest, tachycardia, bradycardia, arrhythmia, fever, and falls).

Develop performance measures for detection accuracy, latency, battery life, and communication integrity.

Select appropriate hardware and software platforms with ROS 2 compatibility for seamless integration into A.L.T.R.U.S.

3.2 Hardware Development for Multi-Sensor

Sensor Selection:

- ECG sensor module for arrhythmia and cardiac event detection.
- Photoplethysmography (PPG) sensor for heart rate and SpO_2 monitoring.
- Digital temperature sensor for core body temperature measurement.
- 3-axis accelerometer and gyroscope for fall detection and activity classification.

Circuit and Enclosure Design:

- Design a low-power, compact-sized PCB to fit all sensors.
- Use an ESP32-based microcontroller with BLE/Wi-Fi for data transmission.
- Design an ergonomic, skin-contact-friendly enclosure for daily use

3.3 Data Acquisition and Preprocessing

Controlled Environment Testing

- Record baseline data sets for all critical signs in controlled resting, walking, and simulated emergency states.

Real-World Pilot Testing:

- Record data from the participants during real-world everyday living and during artificially created fall and motion states.

Preprocessing Pipeline:

- Reduce noise (e.g., bandpass filter for ECG).
- Normalize and resample the signals for consistency.
- Federate sensor data to increase reliability in the case of signal dropouts.

3.4 Machine Learning Model Development

Model Architecture:

- ECG abnormality detection with hybrid deep learning architectures (e.g., Temporal Convolutional Network + LSTM) trained on local as well as public datasets.
- Fall detection with accelerometer–gyroscope time-series classification via a light CNN or random forest classifier for embedded deployment.
- Fever detection with threshold-based methods enhanced with time-based trend analysis.

Training and Validation:

- Use stratified train-test splits and cross-validation to check the model robustness.
- Conduct hyperparameter tuning for optimal accuracy with low computational costs for embedded applications.

3.5 System Integration with A.L.T.R.U.S.

ROS 2 Communication Layer:

- Create a ROS 2 node on the wristband server interface to send sensor and anomaly detection messages.
- A.L.T.R.U.S. will receive said messages and trigger respective actions (alarms, voice commands, emergency calls).

Emergency Response Logic:

- Implement a priority-based action flow:
- Instant local alert to the user.
- Transmission of event data to the robot for validation.
- Activation of emergency procedures in case of non-responsiveness of the user.

Fail-Safe Capabilities:

- Automatic fallback to direct cloud notification in case of failure to communicate with the robot.

3.6 Analysis and Performance Metrics

Accuracy Metrics

- Precision, recall, and F1-score for each type of emergency identified.

Latency

- Duration between event occurrence and notification to robot/system.

Battery Life

- Average runtime under normal usage conditions.

Robustness Testing

- Handling under various lighting, occlusion, and user activity conditions

4. PROJECT REQUIREMENTS

4.1 Functional requirements

Real time physiological monitoring

- Monitor continuously heart rate (HR), oxygen saturation (SpO_2), body temperature, and motion/activity.
- Identify abnormal patterns for:
    - Heart attack
    - Tachycardia
    - Bradycardia
    - Cardiac arrest
    - Falls
    - Fever

Emergency Detection Algorithms

- Employ thresholds and machine learning algorithms to identify life-critical events.
- Provide multi-sensor fusion (ECG, PPG, accelerometer, temperature) for higher accuracy.
- Reduce false positives by considering context (e.g., motion patterns for exercise versus fall).

Notification System

- Send alerts to user and/or caregiver on detected emergencies.
- Send event type, timestamp, and minimal physiological information in alerts.

Data Logging

- Store vital signs over time for trend analysis.
- Store detected emergency events with timestamp and sensor values.
- Support additional analysis by A.L.T.R.U.S. modules.

User Interaction

- Provide minimal on-device feedback (e.g., LED or vibration) on detected anomalies.
- Optional display of current HR, SpO_2, and temperature.

4.2 Non-Functional requirements

Accuracy

- Accuracy of heart rate, SpO_2, and temperature measurement to within clinically acceptable limits.
- Emergency detection algorithms should be directed ≥90% sensitivity and specificity.

Latency

- System should detect and alert emergencies in near real-time (≤5 seconds for critical events like cardiac arrest or fall).

Battery and Power Efficiency

- Run continuously for at least 24 hours on one battery charge.

Wearability and Comfort

- Light and ergonomic design suitable for continuous wear.
- Skin-contact hypoallergenic materials.

Data Security and Privacy

- Ensure safe storage and transmission of confidential health data.
- Comply with research data protection policies that apply (HIPAA/GDPR).

Scalability and Integration

- Must be integrated well with A.L.T.R.U.S.'s main system for further analysis and response.
- Facilitate future integration of sensors or emergency types.

5. BUDGET DETAILS

Item	Estimated Cost (LKR)
Raspberry Pi 5 8GB Original UK	Rs.39,950.00
RPLiDAR 2D	Rs.37,900.00
PK-910H 1080p Full-HD WebCam	Rs.8,000.00
Wheels	Rs.5,000.00
Server motor 2 (Mg90s)	Rs.1,600.00
Gear motor 2 (L298)	Rs.6,000.00
Motor controller	Rs.500.00
ESB 32	Rs.1,700.00
Oled display	Rs.600.00
Heart rate + blood oxygen	Rs.600.00
Ecg	Rs.1,600.00
Accelerometer + gyroscope	Rs.1,500.00
body temperature	Rs.2,500.00
lipo battery	Rs.2,500.00
Gas sensor + Fire sensor	Rs.1,000.00
Additional Cost	Rs.30,000
Total Cost	Rs.140,950.00

Table 1.1 Budget for the proposed system

6. CONCLUSION

The development of the A.L.T.R.U.S. wristband module is a significant step towards proactive and real-time health monitoring. Through the integration of multi-sensor data—such as heart rate, oxygen saturation, body temperature, and movement—the system can accurately detect life-threatening emergencies such as heart attacks, tachycardia, bradycardia, cardiac arrest, falls, and fever. This capability addresses an essential gap in current companion robots, which often depend on inaccurate physiological monitoring and delayed emergency detection.

Through the incorporation of top-level algorithms and real-time notifications, the wristband not only enhances patient safety but also aids caregivers and healthcare providers with decision-making data. Moreover, its wearable nature and storage capacity allow for continuous monitoring without the sacrifice of user convenience. [4]

Overall, this module is contributing towards the greater goal of A.L.T.R.U.S.—an intelligent, responsive, and tailored healthcare companion. Future work can focus on improving the accuracy of predictions, adding more sensors, and broadening its applicability to more general types of health conditions, further rooting it in contemporary, technology-driven healthcare.

References

[1] B. P. P. V. S. F. Davide Gabrielli, "AI on the Pulse: Real-Time Health Anomaly Detection with Wearable and Ambient Intelligence," Ambient Intelligence, New York, 2025.

[2] ,. M. D. I. A. S. K. M. G. Mehmet Yamac ¸ , "A Personalized Zero-Shot ECG Arrhythmia Monitoring System:From Sparse Representation Based Domain Adaption to Energy Efficient Abnormal Beat Detection for Practical ECG Surveillance," 2024.

[3] R. T. R. V.-R. C. M. Juan Carlos Ruiz-Garcia, "CareFall: Automatic Fall Detection through Wearable Devices and AI Methods," 2023.

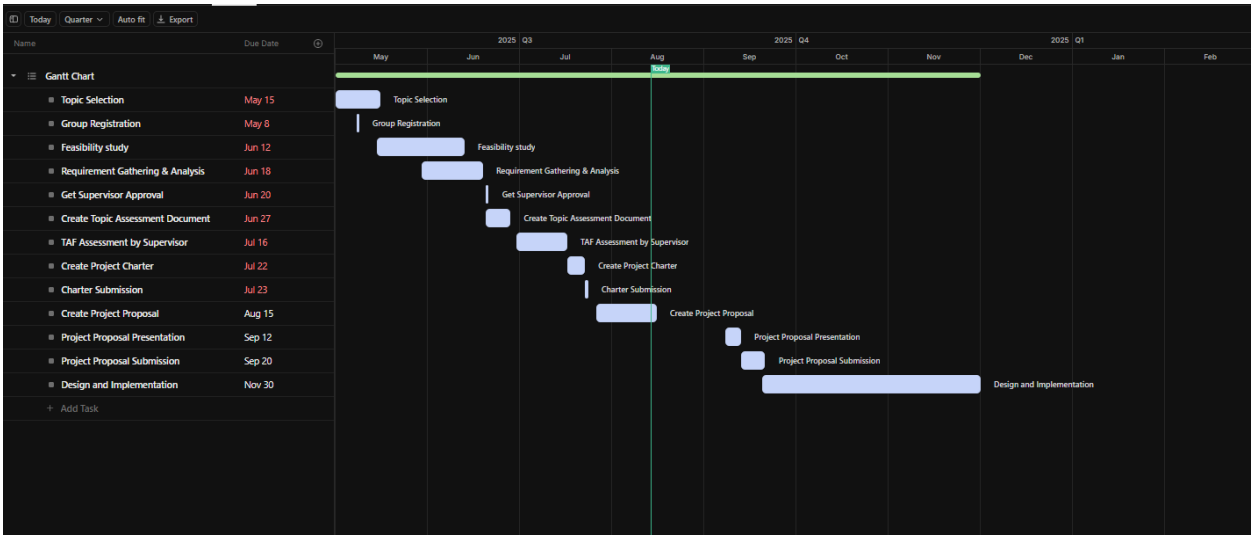
[4] M. O. a. M. H. Saeed Saadatnejad, "LSTM-Based ECG Classification for Continuous Monitoring on Personal Wearable Devices," *Biomedical and Health Informatics (JBHI),* p. 9, 2019.

[5] X. W. M. H. A. B. L. C. L. B. Thorir Mar Ingolfsson, "ECG-TCN: Wearable Cardiac Arrhythmia Detection with a Temporal Convolutional Network," 2021.

APPENDICES

Appendix 1. Work Breakdown Structure (WBS)

Appendix 2. Gantt Chart



Appendix 3. Online Survey

Welcome to the A.L.T.R.U.S Assistive Robot Survey

Thank you for taking the time to participate in our research survey!

About Our Research

Our project is titled **"A.L.T.R.U.S" – Adaptive Life-support & Therapeutic Robotic Unit System**. We are developing an advanced assistive robot designed to support home healthcare by providing safe, intelligent, and empathetic assistance to users with medical or mobility needs.

The goal of A.L.T.R.U.S is to create a robust robot that can:

- Understand and respond to voice and gesture commands naturally
- Monitor health and environmental conditions in real time
- Recognize user emotions and respond empathetically
- Recover from faults and update itself securely

The Four Core Components of A.L.T.R.U.S

Voice-Based Command Interpretation & Adaptive Navigation : The robot understands voice commands and uses smart navigation to move safely around obstacles and adapt to changes in the home environment.
Health and Environmental Monitoring Using Sensor Fusion : Multiple sensors work together to continuously monitor vital signs (like heart rate and temperature) and environmental factors (like gas leaks or air quality) for early emergency detection.
Emotion Detection & Empathetic Interaction : The robot recognizes emotions from facial expressions and voice tone, combining these inputs to respond with empathetic speech and appropriate gestures.
Fault-Tolerant Middleware, Secure Updates & Blockchain Logging : A fault-tolerant system ensures continuous operation even during errors, securely manages software updates, and maintains tamper-proof logs of the robot's activities.

Your feedback is crucial to help us understand which features matter most to potential users. The survey will take about 10 minutes to complete.

Thank you for contributing to this important research!

Experience with Robots or Smart Devices: *

- ○ None
- ○ Basic (e.g., using smart assistants like Alexa/Siri)
- ○ Moderate (e.g., programming robots, using advanced devices)
- ○ High (e.g., robotics development or engineering)

How important is it for the robot to move safely even if the room layout changes (e.g., furniture moved)? *

- ○ Extremely important – safety comes first
- ○ Important – smooth movement matters
- ○ Somewhat important – small mistakes are okay
- ○ Not important – I can guide it myself

What would worry you most about using a robot that moves on its own?" *

- ☐ Bumping into people or objects
- ☐ Not understanding commands
- ☐ Moving to the wrong place
- ☐ Being too slow to react
- ☐ Needing frequent recharging

If the robot could understand both your voice and gestures, would that make it easier for you to use? *

- ○ Yes, much easier
- ○ Yes, a little easier
- ○ No difference
- ○ No, I prefer only one control method

Which features would you like most in a home robot with voice and gesture control? *

- ☐ Understand my voice commands accurately
- ☐ Recognize my hand gestures quickly
- ☐ Avoid obstacles without my help
- ☐ Adapt if the furniture is moved
- ☐ Come to me when I call it

What concerns would you have about using a multi-emergency detection wristband? *

- [ ] Accuracy of health data
- [ ] Privacy of personal health information
- [ ] Battery life
- [ ] Comfort and wearability
- [ ] Cost of the device
- [ ] Risk of false alarms

Have you ever interacted with a robot or virtual assistant? *

- ( ) Yes
- ( ) No

How would you rate your comfort level interacting with robots? *

- ( ) Very comfortable
- ( ) Somewhat comfortable
- ( ) Neutral
- ( ) Uncomfortable
- ( ) Very uncomfortable

Which emotions do you think are most important for a robot to detect? *

- [ ] Happiness
- [ ] Sadness
- [ ] Anger
- [ ] Fear
- [ ] Surprise
- [ ] Disgust

In your opinion, which medical emergencies should a wearable health monitoring wristband prioritize detecting? *

Short-answer text

In your opinion, which medical emergencies should a wearable health monitoring wristband prioritize detecting? *

- ( ) Very familiar, I use one regularly
- ( ) Somewhat familiar, I've heard about them but never used one
- ( ) Not familiar at all

If a wristband alerted you about a possible medical emergency, how likely are you to act immediately (e.g., call emergency services, inform family)? *

- ( ) Very likely
- ( ) Likely
- ( ) Neutral
- ( ) Unlikely
- ( ) Very unlikely

Which of these features would you find most useful in a health emergency detection wristband? *

- [ ] Heart attack detection
- [ ] Tachycardia & bradycardia detection
- [ ] Cardiac arrest detection
- [ ] Fall detection
- [ ] Fever detection
- [ ] Emergency call/SMS alerts
- [ ] Historical health trend analysis

Which of these would you want in a helpful robot? *

- [ ] Understand my requests correctly
- [ ] Fix itself when something goes wrong
- [ ] Keep a secure record of actions
- [ ] Connect different robot features smoothly (e.g., movement + health checks)
- [ ] Share information instantly between parts

What would worry you about using a robot that makes its own decisions? *

- [ ] Safety during malfunctions
- [ ] Misunderstanding commands
- [ ] Privacy of personal information
- [ ] Relying too much on technology
- [ ] Cost of buying or maintaining it
- [ ] Being too complicated to use

From a robotics systems perspective, what core services should middleware guarantee in a mission-critical assistive robot? *

Short-answer text

In modular assistive robots, where would you rank fault tolerance in the middleware priority list? *

- ( ) Top priority – essential for safe autonomy
- ( ) High priority – crucial for user trust
- ( ) Medium priority – nice to have but not critical
- ( ) Low priority – can be handled at application level

How effective do you think blockchain is for ensuring data integrity and accountability in distributed robotic systems? *

- ( ) Very effective – ideal for immutable logging
- ( ) Effective – works well with limitations
- ( ) Neutral – other methods are equally good
- ( ) Ineffective – adds unnecessary complexity

Which features would you expect from a next-generation robot middleware? *

- [ ] Intent recognition and context mapping
- [ ] Real-time fault detection and self-healing
- [ ] Blockchain-based action/event logging
- [ ] Modular plugin management for robot capabilities
- [ ] Developer tools for monitoring and debugging

What challenges do you foresee in implementing fault-tolerant, blockchain-backed middleware for modular robots? *

- [ ] Real-time performance overhead
- [ ] High resource consumption (CPU, storage)
- [ ] Complexity of fault recovery algorithms
- [ ] Blockchain scalability and latency
- [ ] Interoperability between heterogeneous modules
- [ ] Developer adoption and learning curve

Which emotional cues do you think a robot should detect? *

- [ ] Facial expression
- [ ] Tone of voice
- [ ] Gestures/body language
- [ ] Words used in speech

Would you trust a robot that can read your emotions? *

- ( ) Yes
- ( ) Maybe
- ( ) No

If a robot was helping you at home, what tasks would you want it to do without ever failing? *

Short-answer text

How important is it for an assistive robot to keep working even if some parts stop working? *

- ( ) Extremely important – safety comes first
- ( ) Important – reliability matters
- ( ) Somewhat important – minor issues are okay
- ( ) Not important – I can manage without it

If a robot kept a secure, unchangeable record of what it did, would this make you trust it more? *

- ( ) Yes, a lot more
- ( ) Yes, a little more
- ( ) No difference
- ( ) No, it wouldn't matter